



Physics Unit 7

Density aur Temperature

Aasan Notes, tumhari kitab ke saath saath (book pages 167 to 188)

Assalam o alaikum Seemab! Ye chapter abhi class mein nahin parhaya gaya, is liye ye notes tumhari kitab ke bilkul saath chalte hain: wahi headings aur wahi numbers (7.1, 7.2 ...). Kitab saath khol lo to aur bhi aasan lagega.

Kaise parhna hai (sirf 35 se 40 minute):

1. Ek dafa aaram se poore notes parho. Ratna nahin, bas samajhna hai.
2. Phir sirf **neelay**  **boxes** dobara parho. Yehi definitions exam mein likhni hain, kitab ke English alfaaz mein.
3. Aakhir mein "**Khud ko check karo**" ke 12 sawal karo. 9 ya zyada sahi? Shabaash, tum paper ke liye tayyar ho!

Is chapter ka naqsha (5 hisse)

 Hissa 1	Density aur usay naapna (7.1, 7.2)
 Hissa 2	Matter ki states aur Plasma (7.3, 7.4)
 Hissa 3	Temperature, particles aur internal energy (7.5, 7.6)
 Hissa 4	Thermometer kaise kaam karte hain, fixed points (7.8, 7.9)
 Hissa 5	Achha thermometer: sensitivity, range, linearity (7.10 to 7.12)

Note: kitab mein 7.7 number ka koi hissa nahin hai, 7.6 ke baad seedha 7.8 aata hai. Ye kitab ki numbering hai, tumse kuch nahin chhoota.



Hissa 1: Density

7.1 Density (kitab p168)

Ek sawal socho: **1 kilo rui (cotton) zyada jagah gherti hai ya 1 kilo loha?** Wazan dono ka barabar hai, lekin rui ka bara sa dher hota hai aur loha ek chhota sa tukra. Kyun? Kyunke loha mein atoms bohat qareeb qareeb bhare hote hain (**closely packed**). Rui mein beech mein bohat khaali jagah hoti hai.

Bas yehi density hai: **kisi cheez ki ek fixed jagah (volume) mein kitna maal (matter) bhara hua hai.**



Kitab ki definition: exam mein yehi likho

"Mass per unit volume of the substance is called its density."

$$\text{Density} = \text{Mass} / \text{Volume} \quad \rho = m / V$$

- ρ ko "**rho**" parhte hain. Ye density ka symbol hai.
- **SI unit: kg/m^3** (kilogram per cubic metre). Chhota unit: **g/cm^3** . Liquids ke liye aksar **g/mL** .
- Density ek **scalar quantity** hai (sirf number aur unit, koi direction nahin).
- **Solids > Liquids > Gases**. Solids mein atoms sab se qareeb, gases mein sab se door. Is liye solids sab se dense aur gases sab se kam dense.



Unit badalne ki trick (kitab ka FYI box)

$1 \text{ g/cm}^3 = 1000 \text{ kg/m}^3$. g/cm^3 se kg/m^3 banana ho to **1000 se multiply** karo.

Misal: $0.9 \text{ g/cm}^3 \times 1000 = 900 \text{ kg/m}^3$. Aur $1 \text{ mL} = 1 \text{ cm}^3$, is liye g/mL aur g/cm^3 ek hi baat hai.

Table 7.1: kuch cheezon ki density (bhaari se halki tak, kg/m³):

Material	Density (kg/m ³)	Material	Density (kg/m ³)
Gold (sona)	19300	Ice (barf)	920
Mercury (paara)	13600	Polythene	900
Iron (loha)	7900	Petrol	800
Pure water	1000	Carbon dioxide	2.0
		Air (hawa)	1.3

Roz ki misal

Glass mein barf ka tukra hamesha upar kyun tairta (float) hai? Table dekho: **barf 920, paani 1000**. Barf paani se halki (kam dense) hai, is liye upar rehti hai. Aur carbon dioxide (2.0) hawa (1.3) se bhaari hai.

★ MCQ wali baatein

- **Osmium** sab se zyada dense material hai (room temperature aur pressure par): **22.59 g/cm³**. Ye **diamond se bhi sakht (harder)** hai.
- Kisi cheez ka chhota tukra lo ya bara, **density same** rehti hai, kyunke density "unit volume" ka maal hai.
- Temperature badle to density bhi badal jati hai (garam cheez phail jati hai, volume barh jata hai).
- **Liquids gases se zyada dense kyun?** Liquids mein particles qareeb hain; gases mein door door, is liye fixed volume mein maal kam.

7.2 Density kaise maloom karein (kitab p169 to 173)

Zaroori

Hamare paas kitab ke **page 170 aur 171** ki scan nahin hai. Ye do page **apni kitab se zaroor parh lena**, un mein density naapne ke tareeqe ka shuru wala hissa hai.

7.2.1 Regular shape (seedhi, pakki shakal, jaise cube ya eent): volume ruler se naap kar nikal lete hain.

- **Cube ka volume = (side length)³**
- **Brick (eent) ka volume = length × width × height**
- Phir $\rho = m / V$

 Example 7.1 (kitab p169): cube ki density

Cube ki side 5 cm, mass 500 g. Density?

Given: $m = 500 \text{ g}$, side = 5 cm

Required: $\rho = ?$

Solution: $V = (\text{side})^3 = (5 \text{ cm})^3 = 125 \text{ cm}^3$

$$\rho = m / V = 500 \text{ g} / 125 \text{ cm}^3$$

$$\rho = 4 \text{ g/cm}^3$$

 Example 7.3 (kitab p172): eent (brick) ki density

Mass 3.30 kg; length 21.6 cm, width 10.2 cm, height 6.35 cm.

Step 1: cm ko metre mein: 0.216 m, 0.102 m, 0.0635 m

Step 2: $V = 0.216 \times 0.102 \times 0.0635 = 0.00140 \text{ m}^3$

Step 3: $\rho = 3.3 \text{ kg} / 0.00140 \text{ m}^3$

$$\rho \approx 2360 \text{ kg/m}^3$$

 Units milao!

Mass **kg** mein hai to volume **m³** mein lo, jawab **kg/m³**. Mass **g** mein hai to volume **cm³** mein, jawab **g/cm³**. Kabhi kg ko cm³ se divide mat karna.

7.2.2 Irregular shape (tedhi meydhi shakal, jaise pathar): **displacement method.**

 Pyaase kawwe ki kahani yaad hai?

Kawwe ne ghade mein kankar daale aur paani upar aa gaya. Har kankar ne **apne volume jitna paani upar dhakel diya** (displace kiya).

Displacement method bilkul yehi hai! (Ye kahani sirf samajhne ke liye hai, exam mein kitab ke steps likhne hain.)

Kitab ke 6 steps (Figure 7.3):

1. Pathar ka **mass** balance se naapo.
2. Measuring cylinder mein kuch paani daalo, **initial volume (Vi)** note karo.
3. Pathar ko **dhaage (thread)** se baandh kar aahista se cylinder mein utaaro.
4. Paani upar charh jayega (displace hoga). **Final volume (Vf)** note karo. $V_f = \text{paani ka volume} + \text{pathar ka volume}$.
5. **Pathar ka volume = $V_f - V_i$**
6. **Density = mass / volume**

Yaad rakho: ye method un cheezon ke liye hai jo paani mein **doob jayen** (sink) aur **ghulein nahin** (insoluble).

Example 7.4 (kitab p173): pathar ki density

Rock ka mass 80.52 g. Paani 18 mL se 46 mL par aa gaya.

Volume: $V = V_f - V_i = 46 \text{ mL} - 18 \text{ mL} = 28 \text{ mL} = 28 \text{ cm}^3$

Density: $\rho = 80.52 \text{ g} / 28 \text{ cm}^3$

$$\rho = 2.88 \text{ g/cm}^3$$

Khud karo: kitab ke numericals (p188), jawab ke saath

1. Wooden block 300 g, 150 cm^3 aur metal ball 500 g, 50 cm^3 . **2 g/cm^3 aur 10 g/cm^3 ; metal ball zyada dense.**
2. Cooking oil 500 mL, mass 450 g. **0.9 g/mL** (paani 1 g/mL se kam, is liye salan mein tel upar tairta hai!)
3. Plastic box $70 \text{ cm} \times 10 \text{ cm} \times 30 \text{ cm}$, mass 2500 g. **$V = 21000 \text{ cm}^3$, $\rho \approx 0.12 \text{ g/cm}^3$**
6. Pathar 200 g, paani 40 mL se 73 mL. **$V = 33 \text{ mL}$, $\rho \approx 6.1 \text{ g/mL}$**

Q4 aur Q5 abhi chhor do: Q4 ke liye sphere ka formula chahiye jo scan ke missing pages par hai, aur Q5 ka kitab wala jawab Table 7.1 ki iron density se mail nahin khata.



Hissa 2: Matter ki states

7.3 States of matter (kitab p174, 175)

Har cheez chhote chhote **particles** (atoms aur molecules) se bani hai. Ye particles ek doosre ko kheenchte hain (**attractive force**). Ye force kitni mazboot hai, isi se faisla hota hai ke cheez solid hogi, liquid ya gas. Force mazboot = particles qareeb; force kamzor = particles door.



School ki misal (samajhne ke liye)

Solid = assembly ki line mein khari larkiyen. Apni jagah se hil nahin saktin, bas wahin thora sa hilti julti (**vibrate**) hain.

Liquid = break mein corridor. Sab qareeb qareeb hain, lekin ek doosre ke paas se guzar jati hain (**glide**).

Gas = chhutti ke baad ground. Sab door door, tezi se har taraf bhaag rahi hain aur takra rahi hain.

Table 7.2: States of matter (ye poori table yaad karo):

	Solid	Liquid	Gas
Density	High	Medium	Low
Arrangement	Fixed positions	Random positions	Random positions
Movement	Vibrate around a fixed position	Move around each other	Move quickly in all directions
Energy	Low energy	Greater energy	Highest energy

Kitab ke points, har state ke:

- **Solids (7.3.1):** strong attractive force; particles closely packed, **fixed lattice pattern**; na gas ki tarah azad ghoom sakte na liquid ki tarah phisal sakte; sirf apni **mean position** ke gird vibrate karte hain; **fix shape**; high density.
- **Liquids:** force gases se zyada lekin solids se kam (**intermediate**); faasla solids se zyada lekin phir bhi qareeb; particles **flow and glide** over each

other; koi fix shape nahin, **bartan (container) ki shakal** le leta hai; intermediate density.

- **Gases: negligibly weak** force; particles ka faasla un ke apne size se bohat zyada; **constant random motion**, aapas mein aur container ki deewaron se takrate rehte hain; force sirf takrane (collision) ke waqt; **na fix shape na fix volume**, poore container mein phail jati hai (volume = container ka volume); sab se kam dense.

★ **Point to ponder (kitab p176)**

Liquids aur gases ki koi fixed shape nahin, dono beh (flow) sakte hain. Is liye dono ko mila kar **fluids** kehte hain.

Trick: FLUID = jo Flow kare = Liquid + Gas. (Solid fluid nahin hai!)

7.4 Plasma: chauthi (fourth) state (kitab p176, 177)

📖 **Kitab ki baat: exam mein yehi likho**

“Plasma consists of positive ions, free electrons (negatively charged particles) and neutral atoms in gaseous state.”

Plasma banta kaise hai? Gas ko garam karte jao. Particles ki K.E. barhti jati hai, attractive force kam hoti jati hai, aur particles zor zor se takrate hain. Takkar se atoms ke **electrons alag ho jate hain** aur atom **positive ion** ban jata hai. Is **ionic state** ko plasma kehte hain. Free electrons aur moving ions ki wajah se plasma **current bohat achhi tarah conduct** karta hai.

- Aksar **bohat high temperature**, ya **high pressure**, ya dono par banta hai. **High electric aur magnetic field** se bhi ban sakta hai.
- Plasma = gas jo **hot, bright aur highly ionized** ho.
- **99% visible universe** plasma se bana hai.

⚡ Plasma kahan milta hai (kitab ki misalein)

Sun (stars plasma ki wajah se chamakte hain), **neon light**, **lightning** (aasmani bijli), **aurora** (north aur south poles par), **nebula**, **welding arc**, **plasma lamps**.

Gali ke welding wale mistri ki neeli chamakti chingari dekhi hai? Wo welding arc plasma hai!

★ For your information (kitab p177)

Aurora (polar lights): Sun ke charged particles upper atmosphere se takrate hain to aasman mein rangeen roshniyan nazar aati hain.

Nebula: stars ke darmiyan dust aur gas ka bohat bara badal. Latin lafz, matlab "mist, fog". Is mein hydrogen aur ionized gases (plasma) hoti hain; andar ke garam stars inhein garam karte hain to laal, neeli, hari roshni nikalti hai.

🧠 4 states yaad karne ki trick

Solid, Liquid, Gas, Plasma = "Sab Log Garam Pakore" 😊

Aur yaad rakho: Plasma = **ionized state of matter**.

🔥 Hissa 3: Temperature aur particles

7.5 Particles ki motion aur temperature (kitab p177, 178)

📖 Kitab ki misal

Pateele mein paani ke molecules hamesha hil rahe hote hain, lekin nazar nahin aate. Choolha jalao to paani garam ho kar us mein **harkat (currents)** nazar aane lagti hai. Garmi ne molecules ko tez kar diya!

Kisi cheez ko garam karo to **do mein se ek** kaam hota hai:

1. Particles ke darmiyan attractive force kamzor ho kar **bonds toot** sakte hain (jaise **melting** aur **boiling** mein), ya
2. Particles **tez** ho jate hain, yani un ki **K.E. barh jati hai**.

 Kitab ki definition: exam mein yehi likho

"The temperature of a substance is the measure of its hotness or coldness, and the temperature of a substance is directly proportional to the average K.E. of its particles."

Matlab: **particles jitne tez, cheez utni garam.** Garmi do: particles tez, temperature upar. Garmi nikalo (barf par rakho ya fridge mein): particles sust, temperature neeche.

Absolute zero: garmi nikalte jao to particles sust hote hote ek aisi haalat mein aa jate hain jahan un ke paas **least ya no K.E.** hoti hai. Na aapas mein takrate hain na container se, is liye **pressure = 0 Pa.**

 Kitab ki definition

"Absolute zero is the lowest possible temperature of a substance at which its particles have least kinetic energy."

Absolute zero = 0 K = -273.15°C

Is se zyada thanda kuch ho hi nahin sakta. Absolute zero par particles ko hilane ke liye koi heat energy nahin hoti.

Figure 7.8 ke do graphs: (a) Kelvin temperature barhe to particles ki K.E. bhi barhe: **seedhi line** (straight line). (b) Gas ka pressure aur $^{\circ}\text{C}$: line ko peeche barhao to **-273°C par pressure zero** ho jata hai. Yehi absolute zero hai.

 Can you tell? (kitab p183): bohat pasandeeda sawal

Melting point aur boiling point par garmi dene ke bawajood temperature kyun nahin barhta?

Kyunke us waqt garmi particles ke darmiyan **faasla barhane** (force kamzor karne) mein lagti hai taa ke **state badle**. Ye garmi K.E. barhane mein nahin lagti. K.E. nahin barhi to temperature bhi **same** rehta hai.

 Kitab ki printing ghalti

Is box ki aakhri line mein "temperature ... **does change**" chhapa hai. Asal mein "**does not change**" hai. Temperature melting aur boiling point par **nahin badalta**.

7.6 Internal energy (kitab p178, 179)

 **Kitab ki definition: exam mein yehi likho**

"Internal energy is the total kinetic and potential energy of the particles of a substance."

- **Potential energy** particles ke darmiyan **attractive force** ki wajah se hoti hai.
- **K.E. teen qism ki: translational** (seedha chalna), **rotational** (ghoomna), **vibrational** (thartharana).
- **Ideal gas** mein sirf translational K.E. hoti hai, is liye us ki internal energy **sirf K.E.** ki wajah se hai.

 **SLO wali line: zaroor yaad karo**

"By increasing the temperature of a substance, its internal energy also increases."

 **Misal (samajhne ke liye)**

Poori class ki "total pocket money" = har larki ke paise jama karo. Isi tarah **internal energy = har particle ki K.E. + P.E. ka total**. Garam karne se har particle ki K.E. barhi, to total bhi barh gaya.

 **Teeno saath chalte hain**

Temperature $\uparrow$ $\rightarrow$ K.E. $\uparrow$ $\rightarrow$ Internal energy $\uparrow$

Internal energy barhne ka matlab hai temperature barha. Ye garmi dene se bhi ho sakta hai aur kisi aur tareeqe se bhi (kitab poochti hai: socho, haath zor se ragro to kya hota hai?).



Hissa 4: Thermometer kaise kaam karte hain

7.8 Physical properties se temperature naapna (kitab p179, 180)

Garmi se particles tez hote hain, aur cheezon ki kuch **physical properties** ek pakke, pehle se maloom andaz (**predictable way**) mein badalti hain: **volume, pressure, colour, electrical resistance**. Inhi tabdeeliyon ko dekh kar temperature naapte hain.



Kitab ki definition

The physical property on the basis of which a thermometer works is called its **thermometric property**.

7.8.1 Liquids ka phailna (expansion): zyada tar liquids garam ho kar phailte hain. Jo liquid **uniformly** (barabar) phailta hai woh thermometric material ban sakta hai: **mercury ya alcohol**. Thermometer garam cheez ko chhuye to mercury phailti hai, tube mein upar charhti hai, aur scale par reading aa jati hai. Mercury ka phailna temperature ke saath **linearly** badalta hai.



Roz ki misal

Ammi jab bukhari check karti hai to thermometer mein chamakdaar line upar charhti hai. Wo mercury ka phailna hai!

7.8.2 Gas ka volume aur pressure: gas garam ho to us ka volume aur pressure badal sakte hain. Gas thermometer **do qism** ke:

- **Constant volume gas thermometer:** volume same rakhte hain; **pressure** temperature ke saath directly proportional. Law: **Gay-Lussac's law**. Garam bulb mein gas ke particles zor se takrate hain, pressure barhta hai.
- **Constant pressure gas thermometer:** pressure same rakhte hain; **volume** temperature ke saath proportional. Law: **Charles's law**. Gas phail kar **piston** ko dhakelti hai.

Trick

"Charles ke paas PC hai": Pressure Constant = Charles's law.

Baqi bacha: Volume constant = Gay-Lussac's law.

7.8.3 Liquid crystals ka rang: liquid crystal ek **plastic strip ya patch** mein band hota hai. Jism se lagao, temperature badle to **rang (colour) badalta** hai. Rang ko temperature scale se milao aur temperature maloom. Use: **fever thermometers, aquariums, baby bottles** (kitab mein bache ke maathe par strip ki tasveer hai).

7.8.4 Resistance thermometer / thermistor: temperature badalne se is ki **resistance** (current ke raaste mein rukawat) badalti hai.

7.8.5 Bimetallic strip thermometer: solid (do metals ki patti) ke phailne se kaam karta hai.

Kitab ki ghalti (p180)

7.8.5 ki line mein bimetallic strip aur thermocouple ki baat ghalti se aapas mein mil gayi hai. **Neeche wali Table 7.3 ka jora yaad karo, woh sahi hai.**

Table 7.3: thermometer aur us ki thermometric property (MCQs ke liye bohat zaroori):

Thermometric property	Thermometer
Volume expansion of a gas	Constant pressure gas thermometer
Volume expansion of a liquid	Laboratory or clinical thermometer
Volume expansion of a solid	Bi-metallic strip thermometer
Pressure change of a fixed mass of gas	Constant-volume gas thermometer
Changes in e.m.f	Thermocouple
Changes in electrical resistance	Resistance thermometer or thermistor

Ulta jora yaad rakho

Gas ka **volume** naap rahe ho? To **pressure** constant hai. Gas ka **pressure** naap rahe ho? To **volume** constant hai. Jo cheez naapte ho, woh **constant nahin** ho sakti!

Solid phaile = **bimetallic**. **e.m.f / voltage** = **thermocouple**.

7.9 Fixed points (calibration) (kitab p181, 182)

Socho tumhare paas ek ruler hai jis par koi nishan (marking) nahin. Kya us se kuch naap sakti ho? Nahin! Thermometer ka bhi yehi haal hai: **scale ke baghair thermometer "valueless" hai**. Theek scale banane ke liye **do fixed points** chahiye: neeche wala aur upar wala. Phir un ke beech ke faasle ko **barabar hisson (equal divisions)** mein baant dete hain.

Kitab ki definition

"A fixed point is a standard degree of hotness or coldness."

- Fixed points ko **reference points** bhi kehte hain: **lower reference point** aur **upper reference point**.
- Fixed points **accessible** (asaani se mil saken) aur **reproducible** (baar baar wahi aayen) hone chahiye.
- Fixed points tay kiye baghair temperature **accurately** nahin naap sakte.
- **Lower** = barf ke pighalne ka point (**ice point**). **Upper** = paani ka boiling point (**steam point**).

Scale	Ice point (lower)	Steam point (upper)	Divisions
Celsius	0 °C	100 °C	100
Fahrenheit	32 °F	212 °F	180
Kelvin	273 K	373 K	100

Trick

Divisions = upper – lower. ($100 - 0 = 100$; $212 - 32 = 180$; $373 - 273 = 100$)

Kelvin = Celsius + 273. Fahrenheit hi akela "ajeeb" hai: **32 se shuru, 212 par khatam, 180 hisse.**



Hissa 5: Achha thermometer kaisa hota hai

7.10 Sensitivity, range aur linearity (kitab p182, 183)

7.10.1 Sensitivity

Kitab ki definition

"It is ability of a thermometer to detect small changes in temperature."

- 0.1 °C sensitivity wala thermometer 0.1 °C ki chhoti si tabdeeli bhi pakar leta hai. 1 °C wala, 1 °C se chhoti tabdeeli nahin naap sakta.
- Sensitivity **least count** jaisi hai (analogous to least count).
- **Mercury thermometer sensitive** hota hai: room temperature ke paani mein garam paani ke chand qatre daalo, mercury foran upar.



Misal (samajhne ke liye)

Sunar (jeweller) ka taraazu aur sabzi wale ka taraazu. Sunar ka taraazu chhoti se chhoti tabdeeli bhi pakar leta hai, woh zyada "sensitive" hai.

7.10.2 Range

Kitab ki definition

"The range of a thermometer is the range of temperatures that it can measure. It is measured by the lowest and highest temperatures that the thermometer can measure."

- **Clinical mercury thermometer:** 35 °C to 42 °C (Fahrenheit par **94 °F to 108 °F**).
- **Digital thermometer** ki range **wide** hai. Barf mein 0 °C aur ubalte paani mein 100 °C sahi dikhata hai. Isi tarah range check karte hain.

😊 Clinical thermometer sirf insani jism ke liye bana hai (35 se 42). Is se garam chai ka temperature mat naapna!

7.10.3 Linearity

Kitab ki definition

“Linearity tells us how equally and uniformly a thermometer shows the temperature variations.”

- Scale par nishan **barabar faasle** par (equally spaced). 0 se 50 °C ka izafa aur 50 se 100 °C ka izafa, dono **same accuracy** se naape jayen.
- **Bimetallic thermometers linear** hote hain.

Misal

Seerhiyan: har seerhi ki unchai barabar ho to chadhna aasan hai.
Thermometer ki "seerhiyan" (divisions) barabar hon, yehi linearity hai.

7.11 Liquid-in-glass aur thermocouple thermometers (kitab p183, 184)

7.11.1 Liquid-in-glass: lambi, patli glass tube, aakhir mein **bulb**; bulb mein **mercury ya alcohol**. Garam: liquid **phail kar** tube mein upar. Thanda: **sikur kar** neeche. Scale calibrated hota hai, liquid ki unchai se reading. Use: **ghar, school, laboratories**, aur kuch industries.

Figure 7.14: glass thermometer ke parts: mercury bulb, constriction, capillary tube, column of mercury, scale lines, stem, expansion chamber.

7.11.2 Thermocouple: do mukhtalif metals ki taarein ek sire par juri hoti hain: ye **hot junction**. Doosre sire (**cold junctions**) voltmeter se jure hote hain. Hot junction garam ho to metals mein free electrons behte hain aur hot aur cold junction ke darmiyan **voltage** ban jata hai. Zyada garam = zyada voltage; thanda = kam voltage. Temperature ki tabdeeli voltage ki tabdeeli ke

linearly proportional hai. Voltmeter ko calibrate kar dete hain taa ke voltage se temperature parha ja sake.

Use: **furnaces, kilns, engines**, industrial equipment; **agriculture** mein mitti (soil) aur paani ka temperature. (Figure 7.13 mein 829 °C / 1500 °F likha hai.)

	Liquid-in-glass	Thermocouple
Structure	Glass tube + bulb (mercury/alcohol)	2 different metals ki wires: hot junction + cold junctions
Kaam	Liquid ka expansion aur contraction	Voltage (e.m.f) ka farq
Kahan	Ghar, school, lab	Furnaces, kilns, engines, industry, agriculture

7.12 Structure ka asar (kitab p184, 185)

Structure ki cheez	Kis par asar	Kaise
Tube ka diameter	Sensitivity	Patli (narrow) capillary tube = zyada sensitive. Liquid kam, jaldi react kare, jaldi upar charhe.
Liquid ki qism	Sensitivity	Mercury aur alcohol paani se zyada phailte hain = zyada sensitive.
Bulb ka size	Range	Bara bulb = zyada liquid phailne ke liye = large range.
Liquid ki qism	Range	Mercury: melting point kam, boiling point zyada = wide range.
Glass ki qism	Linearity	Borosilicate glass , soda lime glass se zyada linear.

 **Trick: "Nali, Bulb, Glass"**

Patli Nali = Sensitive. Bara Bulb = Range. Borosilicate Glass = Linear.

 Straw socho: patli straw mein thora sa juice bhi kaafi upar charh jata hai. Yehi patli tube ki sensitivity hai.



Summary: kitab ke alfaaz mein yaad karo

- Density is mass per unit volume.
- States of matter are four, which are named as Solids, Liquids, Gases and Plasma.
- Plasma is called fourth state of matter which is the ionized state of matter.
- Thermometry is the branch of physics, which deals with the measurements of temperature.
- Temperature is the measure of degree of hotness or coldness of a body.
- Thermometer is a device which is used to measure temperature.
- Heat is the form of energy which is transferred from one body to another body due to the difference in temperature.

English alfaaz ki chhoti dictionary

closely packed	qareeb qareeb bhare hue	lattice	particles ka pakka jaal/pattern
vibrate	apni jagah thartharana	ionized	electrons alag ho gaye
displace	jagah se hataana	insoluble	jo ghule nahin
calibrate	sahi scale lagana	reproducible	baar baar wahi aaye
capillary tube	baal jaisi patli tube	junction	jahan taarein milti hain

✓ Khud ko check karo (12 sawal)

Pehle khud jawab socho ya notebook mein likho, phir agle page par check karo.

1. Density ka SI unit kya hai?
2. 2.5 g/cm^3 ko kg/m^3 mein likho.
3. Cube ki side 2 cm, mass 40 g. Density kitni?
4. Pathar ka mass 60 g. Paani 30 mL se 50 mL par aa gaya. Pathar ki density?
5. Barf paani par kyun tairti hai? (Table 7.1 ke numbers se batao.)
6. Kis state mein particles sirf apni fixed position ke gird vibrate karte hain?
7. Liquids aur gases ko mila kar kya kehte hain?
8. Absolute zero Celsius mein kitna hai?
9. Constant volume gas thermometer kis law par kaam karta hai?
10. Thermometer ko zyada sensitive banana ho to capillary tube kaisi honi chahiye?
11. Fahrenheit scale par ice point aur steam point ke darmiyan kitne divisions hain?
12. Thermocouple ki thermometric property kya hai?

Agar kuch samajh na aaye (optional)

NBF Physics 9 Unit 7 short questions ki video: youtube.com/watch?v=PHh4-85H4E4

Notes kaafi hain; video sirf tab dekho jab koi hissa mushkil lage.

Jawab

1. **kg/m³** (kilogram per cubic metre)
2. $2.5 \times 1000 = \mathbf{2500 \text{ kg/m}^3}$
3. $V = 2^3 = 8 \text{ cm}^3$; $\rho = 40 / 8 = \mathbf{5 \text{ g/cm}^3}$
4. $V = 50 - 30 = 20 \text{ mL} = 20 \text{ cm}^3$; $\rho = 60 / 20 = \mathbf{3 \text{ g/cm}^3}$
5. Barf ki density **920 kg/m³**, paani ki **1000 kg/m³**. Barf kam dense hai, is liye upar tairti hai.
6. **Solid**
7. **Fluids**
8. **-273.15 °C** (yani 0 K)
9. **Gay-Lussac's law** (pressure temperature ke directly proportional, volume constant)
10. **Patli (narrow / small diameter)**
11. **180 divisions** (212 – 32)
12. **Changes in e.m.f** (voltage)

Score dekho:

- **10 se 12 sahi:** zabardast! 🌟 Tum paper ke liye bilkul tayyar ho.
- **7 se 9 sahi:** bohat achha! Jo ghalat hue, sirf un ka hissa notes mein dobara dekh lo.
- **7 se kam:** koi baat nahin. Notes ek dafa aur aaram se parho, doosri dafa hamesha aasan lagta hai. 💪

Ye chapter class mein abhi nahin hua, is liye agar paper mein kuch mushkil lage to ghabrana nahin. Maqsad sirf ye dekhna hai ke kya dobara samjhana hai.